

Separating Mixtures — Paper A

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, name the **property that differs** between the substances — soluble or insoluble, boiling point, magnetic, particle size — because that property is why the method works.
- The mark for each question is shown in square brackets on the right.
- R_f = distance moved by the substance ÷ distance moved by the solvent. Both from the baseline. No units.

SECTION 1 – MATCHING THE METHOD

- 1.** For each mixture, write the best method to separate it. [5]

Mixture	Method
sand and water	
salt dissolved in water — you want the salt	
salt dissolved in water — you want the water	
iron filings and sand	
the dyes in a black ink	

- 2.** In filtration, give the name for:
- (a)** the solid left on the filter paper. [1]

- (b)** the liquid that passes through. [1]

- 3.** Muddy water is poured through filter paper.
- (a)** State what is left on the paper and what passes through. [1]

(b) Explain, in terms of particle size, why the mud is trapped but the water is not. [2]

SECTION 2 – APPARATUS

4. The diagram below is left for you to draw. Draw and label the apparatus for **filtration**: filter funnel, filter paper, conical flask, residue, filtrate. [4]

5. Describe how you would obtain dry salt crystals from a salt solution, naming the apparatus you would use. [3]

6. A student wants to separate iron filings from sand.

(a) Name the method. [1]

(b) Explain why the method works. [2]

7. Explain why **filtration** cannot be used to remove salt from sea water. [3]

8. Explain why **evaporation** is the wrong method if you want to collect pure water from sea water, and name the right one. [3]

9. State **two** rules for setting up paper chromatography correctly, and give the reason for each. [4]

End of paper. Check every answer has a unit or a form the question actually asked for.